Photosynthesis — Study Notes

Overview

Photosynthesis is the process by which plants, algae and cyanobacteria use light energy to turn carbon dioxide and water into sugar, releasing oxygen as a by-product.

- **Why it matters:** almost all food chains start with sugar made this way, and almost all atmospheric oxygen comes from it.
- **Energy source:** sunlight, which comes from nuclear fusion in the Sun. So nearly all life on Earth is ultimately fusion-powered.
- **Relation to respiration:** photosynthesis is broadly cellular respiration in reverse. Respiration breaks glucose down with O_2 to release energy and CO_2; photosynthesis builds glucose from CO_2 using light energy and releases O_2.
- **Not very efficient:** the process is complex and its key enzyme (RuBisCO) is slow and error-prone, but it has worked well enough to dominate the planet.
- **Two stages:**
    a. **Light-dependent reactions** (thylakoid membranes): capture light energy and store it as ATP and NADPH; split water and release O_2.
    b. **Light-independent reactions / Calvin cycle** (stroma): use ATP and NADPH to fix CO_2 into sugar (G3P).

Raw materials and how a plant gets them

Input	Entry point	Route
Water (H_2O)	Roots	Carried up to the leaves through **xylem** (vascular tissue)
Carbon dioxide (CO_2)	Leaves	Diffuses in through **stomata** (tiny leaf pores); O_2 exits the same way
Light (photons)	Leaves	Absorbed by the pigment **chlorophyll** in chloroplasts

Plants keep internal O_2 levels in leaves low where possible, because O_2 competes with CO_2 at RuBisCO (see Photorespiration).

The chloroplast: where it happens

Photosynthesis takes place in **chloroplasts**, a type of plastid found in plant cells but not animal cells. A chloroplast is not just a bag of chlorophyll; it has a layered internal structure.

Part	What it is	Role in photosynthesis
Thylakoid	Flattened membrane sac	Its membrane holds chlorophyll and the protein complexes of the light reactions
Granum (pl. grana)	A stack of thylakoids	Packs a large membrane area into a small

Part	What it is	Role in photosynthesis
		space
Lumen	Space inside a thylakoid	Protons (H^+) build up here, forming a gradient
Stroma	Fluid around the thylakoids, inside the chloroplast	Site of the Calvin cycle

- Thylakoid membranes are **phospholipid bilayers**, so they can hold a **concentration gradient**: a higher concentration of ions (here H^+) on one side than the other. This gradient is what later drives ATP production.
- **Chlorophyll** absorbs mainly red and blue light and reflects green, which is why leaves look green. Chlorophyll a is the main pigment; chlorophyll b and carotenoids act as accessory pigments that widen the range of light captured.

Stage 1: Light-dependent reactions

The light reactions turn light energy into chemical energy (ATP and NADPH) and release O_2 from water. All four protein complexes involved sit across the **thylakoid membrane**.

Photoexcitation: when a photon strikes chlorophyll, one of its electrons absorbs the energy and jumps to a higher energy level (it becomes "excited"). The rest of this stage is about capturing that energy before it is lost. The series of carriers that passes the electron along, extracting energy at each step, is the **electron transport chain**.

The four complexes, in order

1. **Photosystem II (PSII)** — the *first* complex, despite its name (it was discovered second). A large complex of proteins and pigments with dozens of chlorophyll molecules; its reaction centre is **P680**.
    - Light excites an electron, which is passed to a **mobile electron carrier** (plastoquinone).
    - To replace the lost electron, PSII **splits water** (photolysis): $2H_2O \rightarrow 4H^+ + 4e^- + O_2$.
    - The H^+ stays in the lumen; the **O_2 is released** — the source of the oxygen we breathe.
2. **Cytochrome b_6f complex** — passes electrons from PSII towards PSI and uses some of their energy to **pump more H^+ into the lumen**.
3. **Photosystem I (PSI)** — reaction centre **P700**. By now the electrons have lost much of their energy, so a second photon **re-energises** them. They are handed to another carrier (ferredoxin).
4. **$NADP^+$ reductase** — combines 2 electrons and 1 H^+ with **$NADP^+$** to form **NADPH**, an energy-carrying molecule ($NADP^+$ is derived from vitamin B_3, niacin).

Making ATP: chemiosmosis

- Water splitting and the cytochrome pump fill the lumen with H^+, charging the thylakoid **like a battery**.
- H^+ flows back out to the stroma through the enzyme **ATP synthase**, which uses that flow to add a phosphate to ADP: **ADP + P_i → ATP**.
- Making ATP with light-driven proton flow is called **photophosphorylation**.

- **Non-cyclic** flow (water → PSII → PSI → NADPH) makes both ATP and NADPH. **Cyclic** flow sends electrons from PSI back to the cytochrome complex, making extra ATP only, which balances the ATP:NADPH ratio the Calvin cycle needs.

Outputs of stage 1

Product	Goes to	Purpose
ATP	Calvin cycle	Energy
NADPH	Calvin cycle	Energy + electrons (reducing power)
O_2	Atmosphere	By-product

Stage 2: Light-independent reactions (Calvin cycle)

The Calvin cycle uses ATP and NADPH from stage 1 to build sugar from CO_2. It runs in the **stroma**.

- Sometimes called the **"dark reactions"**, but this is misleading: they normally run in daylight alongside the light reactions. They simply don't use photons directly, so "light-independent" is the accurate term.
- It is a **cycle** because its starting molecule, **RuBP** (ribulose-1,5-bisphosphate, a 5-carbon sugar), is regenerated at the end.

The three phases (per 3 CO_2)

1. **Carbon fixation** — the enzyme **RuBisCO** attaches CO_2 to RuBP. The resulting 6-carbon molecule is unstable and immediately splits into two **3-phosphoglycerate (3-PGA)** molecules, the first stable product. With 3 RuBP + 3 CO_2 this gives 6 × 3-PGA.
2. **Reduction** — ATP adds a phosphate to each 3-PGA, then NADPH donates electrons, producing **glyceraldehyde-3-phosphate (G3P)**, a high-energy 3-carbon sugar. Result: 6 × G3P.
3. **Regeneration** — 5 of the 6 G3P are rearranged (using more ATP) to rebuild the 3 RuBP. Only **1 G3P leaves the cycle** as net product.

The cycle's balance sheet

Cycle turns	CO_2 in	ATP used	NADPH used	Net G3P out
3 turns (3 CO_2)	3	9	6	1
6 turns (6 CO_2)	6	18	12	2 (= 1 glucose)

Why G3P is the key product

G3P is the true end product of photosynthesis: the plant converts it into almost any carbohydrate it needs.

- **Glucose / sucrose** — short-term energy and transport
- **Starch** — long-term energy storage
- **Cellulose** — cell-wall structure

- Also a starting point for amino acids and lipids

RuBisCO and photorespiration

RuBisCO (ribulose-1,5-bisphosphate carboxylase/oxygenase) is the enzyme that fixes carbon in the Calvin cycle, and its flaws shape how plants work.

- **Origin:** RuBisCO evolved in early single-celled photosynthesisers when the atmosphere had plenty of CO_2 and almost no O_2. It converts inorganic carbon (CO_2) into organic carbon.
- **Slow:** it fixes only a few CO_2 molecules per second, so plants compensate by making huge amounts of it. It is considered the most abundant protein on Earth, with a recent estimate of about **0.7 billion tonnes** worldwide (Bar-On & Milo, 2019).
- **Confuses O_2 for CO_2:** as photosynthesis raised atmospheric O_2, RuBisCO began binding O_2 instead of CO_2 (the "oxygenase" part of its name). In hot, dry conditions this can happen in a large share of reactions.

Photorespiration

When RuBisCO adds O_2 to RuBP, it produces **2-phosphoglycolate**, a toxic compound that inhibits Calvin-cycle enzymes.

- The plant must recycle it through extra reactions (in the chloroplast, peroxisome and mitochondrion), yielding the amino acid **glycine** and some compounds that can re-enter the Calvin cycle.
- This costs ATP, releases fixed CO_2, and fixes no new carbon, so it lowers photosynthetic efficiency.
- This is why plants try to keep O_2 low and CO_2 high around RuBisCO.

How some plants reduce photorespiration

Plant type	Strategy	Examples
C3	No special mechanism; first product is 3-carbon 3-PGA	Wheat, rice, most trees
C4	Fixes CO_2 first into a 4-carbon acid in mesophyll cells, then releases it at high concentration around RuBisCO in bundle-sheath cells (separation in **space**)	Maize, sugarcane
CAM	Opens stomata at night to take in and store CO_2 as acids; runs the Calvin cycle by day with stomata closed (separation in **time**)	Cacti, pineapple

Quick revision

Stage 1 vs stage 2

Feature	Light-dependent reactions	Calvin cycle

Feature	Light-dependent reactions	Calvin cycle
Location	Thylakoid membrane	Stroma
Needs light directly?	Yes	No (but runs in daytime)
Inputs	Light, H_2O, ADP, $NADP^+$	CO_2, ATP, NADPH
Outputs	ATP, NADPH, O_2	G3P, ADP, $NADP^+$
Key players	PSII, cytochrome b_6f, PSI, ATP synthase	RuBisCO, RuBP

Key terms

- **Photoexcitation** — an electron gaining energy from absorbed light.
- **Photolysis** — light-driven splitting of water at PSII; the source of O_2.
- **Electron transport chain** — carriers that pass electrons along, capturing their energy.
- **Chemiosmosis** — ATP made by H^+ flowing down its gradient through ATP synthase.
- **Carbon fixation** — attaching CO_2 to an organic molecule (RuBP).
- **G3P** — 3-carbon sugar; the net product used to make glucose, starch and cellulose.
- **Photorespiration** — RuBisCO fixing O_2 instead of CO_2; wasteful.

Common exam traps

- PSII acts **before** PSI.
- The O_2 released comes from **water**, not from CO_2.
- "Dark reactions" do **not** happen only at night.
- The first stable product of C3 fixation is **3-PGA**; the net exported product is **G3P**.

Sources

- Processes of the Light-Dependent Reactions — LibreTexts (Raven Biology)
- Carbon-fixing enzyme 10 times more abundant than previously thought — Chemistry World